

- Q.28 Describe any three properties of engine lubricants.
 Q.29 Explain the concept of reversibility cycle.
 Q.30 Explain the types of air cleaners.
 Q.31 Describe the methods of finding the indicated and brake horse power.
 Q.32 Write down latest standards for emission instituted by GOI.
 Q.33 Explain the different additives used for improving the lubricant properties.
 Q.34 Explain the working of various components used in air intake system.
 Q.35 Write down name of any five parts of IC engine along with their materials.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Describe the working of fuel injection system along with its different components and their functions.
 Q.37 What is the importance of testing of IC Engines? Describe any one test for testing an IC engine in detail.
 Q.38 What are different types of lubrication systems? Explain working of any one lubrication system.

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No. of Printed Pages : 4 180145/120145/030145
 Roll No.

4th Sem / Agri. Engg. Sub : IC Engines/ Farm IC Engines

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The internal diameter of cylinder is called
 a) Bore b) Swept volume
 c) Stroke bore ratio d) Stroke
 Q.2 The thermal efficiency of diesel engines is about
 a) 50% b) 65%
 c) 70% d) 80%
 Q.3 What is the most common firing order of a four stroke I.C. engine?
 a) 1-3-4-2 b) 1-2-3-4
 c) 4-3-1-2 d) 4-3-2-1
 Q.4 Which of the following does not relate to spark ignition engine?
 a) Spark plug b) Carburetor
 c) Fuel injector d) Ignition coil
 Q.5 Compression ratio of a petrol engine lies in the range of
 a) 2 to 4 b) 7 to 10
 c) 14 to 20 d) None of these

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- Q.6 The ratio of the brake power to the indicated power is called _____.
 a) Mechanical efficiency
 b) Overall efficiency
 c) Indicated thermal efficiency
 d) Volumetric efficiency
- Q.7 A thermosyphon cooling system. the water circulates due to
 a) A water pump b) Density differences
 c) Pressure differences d) Engine speed
- Q.8 A diesel engine works on
 a) Otto cycle b) Carnot cycle
 c) Diesel cycle d) Dual cycle
- Q.9 The specific fuel consumption of a diesel engine as compared to that for petrol engine lies in the range of
 a) Lower
 b) Higher
 c) Same for same output
 d) None of these
- Q.10 What are the various methods used for circulating water around the cylinder and cylinder heads?
 a) Thermo-syphon cooling
 b) Forced or pump cooling
 c) Cooling with thermostatic regulator
 d) All of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 State function of thermostat valve.
 Q.12 Define top dead centre.
 Q.13 What is the objective of Lubrication?
 Q.14 The connecting rod converts reciprocating motion to _____.
 Q.15 Write the function of oil ring.
 Q.16 Morse test is used for _____.
 Q.17 The thermostat valve operates at temperature _____.
 Q.18 Give the function of fuel injector.
 Q.19 Define horse power.
 Q.20 Piston compression rings are made of _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working principle of rotary engine.
 Q.22 Write any five differences between four stroke and two stroke engine.
 Q.23 Enlist the preventive maintenance of IC engines.
 Q.24 Explain the exhaust smoke analysis.
 Q.25 Describe the necessities of engine cooling.
 Q.26 State about different types of alternate fuels used in IC engines.
 Q.27 Explain the working of diesel cycle along with diagram.

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